Supplementary Information for: From microgravity boundary-layer biophysics to the spaceflight multi-ome: a coupled lattice-Boltzmann and Farquhar–von Caemmerer–Berry model of *Arabidopsis* gas exchange and enclosure carbon starvation

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Abstract

This document contains Supplementary Tables S1–S11 and Supplementary Figures S1–S8 supporting the main manuscript. It reports: (i) zero-dimensional enclosure and Farquhar–von Caemmerer–Berry (FvCB) model parameterizations (Tables S1–S2); (ii) lattice-Boltzmann grid convergence index (GCI), stationarity verification, density-matched pressure-outlet throughput checks, leaf-shape/curl sensitivity, surface CO_2 distribution percentiles, chamber domain-size sensitivity, and real BRIC/CARA Petri-dish cross-section simulations including humidity-limited transpiration and wet-agar stability bounds (Tables S3–S6, Figs. S1–S7); and (iii) enclosure drawdown sensitivity sweeps, the complete NASA OSDR metabolomics census, the 21-compound predicted metabolite specification, PaintOmics locus-to-pathway Sankey conservation and MapMan reconstruction verification, and metabolite-free pathway enrichment controls (Tables S7–S11, Fig. S8).

Supplementary Tables

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Table S1: **Zero-dimensional (0-D) enclosure model specifications across spaceflight hardware classes.** Parameters governing dynamic headspace mass balance, boundary-layer drawdown, and 12 h photoperiod carbon gain (`validation/feedback_0d.py`, `scripts/enclosure_sensitivity.py`).

Parameter	**BRIC (Sealed)**	**CARA (Micropore tape)**	**VEGGIE (Ventilated)**
Headspace volume V	30 mL (3×10^{-5} m^3)	30 mL (3×10^{-5} m^3)	∞ (open/circulated)
Leaf area A_{leaf}	3 cm^2 (3×10^{-4} m^2)	3 cm^2 (3×10^{-4} m^2)	3 cm^2
Enclosure seal area A_{dish}	28 cm^2 (2.8×10^{-3} m^2)	28 cm^2 (venting perimeter)	N/A
Seal permeance k_{seal}	0.0 mol m^{-2} s^{-1} (hermetic)	0.006 mol m^{-2} s^{-1}	Actively swept to cabin
Molar air density ρ_{air} (25 °C)	41.6 mol m^{-3}	41.6 mol m^{-3}	41.6 mol m^{-3}
Initial headspace CO_2 ($C_a(0)$)	400 μmol mol^{-1} (ppm)	400 μmol mol^{-1}	400 μmol mol^{-1}
Initial headspace O_2	210 mmol mol^{-1} (21%)	210 mmol mol^{-1}	210 mmol mol^{-1}
Wall boundary condition	Zero flux ($\partial C/\partial n = 0$)	Relaxed to cabin (k_{mem})	Pinned at cabin air
Rosette g_{bl} (μg)	0.272 mol m^{-2} s^{-1}	0.272 mol m^{-2} s^{-1}	0.272 mol m^{-2} s^{-1} (unvented ref.)
12 h photoperiod carbon gain	$\approx$ 1% of Earth	$\approx$ 90% of Earth	$\approx$ 100% of Earth

Table S2: **Farquhar–von Caemmerer–Berry (FvCB) C_3 model parameterization at 25 °C.** *In vivo* Rubisco kinetic constants and Arrhenius activation energies from Bernacchi et al. (2001, 2003); photosynthetic capacities and internal conductances are representative *Arabidopsis thaliana* values (`scripts/fvcb.py`).

Symbol	**Biochemical / biophysical quantity**	**Value (25 °C)**	**Units / Source**
$V_{c\,\max,25}$	Maximum Rubisco carboxylation capacity	90.0	μmol m^{-2} s^{-1}
$J_{\max,25}$	Maximum electron transport rate ($\sim 1.7 \times V_{c\,\max}$)	153.0	μmol m^{-2} s^{-1}
$R_{d,25}$	Day (mitochondrial) respiration	1.35	μmol m^{-2} s^{-1}
$K_{c,25}$	Michaelis constant of Rubisco for CO_2	404.9	μmol mol^{-1} (Bernacchi 2001)
$K_{o,25}$	Michaelis constant of Rubisco for O_2	278.4	mmol mol^{-1} (Bernacchi 2001)
Γ^*_{25}	Photorespiratory CO_2 compensation point (21% O_2)	42.75	μmol mol^{-1} (Bernacchi 2001)
$S_{c/o,25}$	Implied Rubisco CO_2/O_2 specificity ($0.5\,O_{\mathrm{ref}}/\Gamma^*_{25}$)	2456.1	mol mol^{-1}
g_s	Stomatal conductance to CO_2	0.20	mol m^{-2} s^{-1}
g_m	Mesophyll conductance to CO_2	0.30	mol m^{-2} s^{-1}
θ	Non-rectangular hyperbola curvature of light response	0.70	Dimensionless
α	Quantum yield of whole-chain electron transport	0.30	mol e$^-$ mol^{-1} photon
$E_a(V_{c\,\max})$	Activation energy of $V_{c\,\max}$	65.33	kJ mol^{-1}
$E_a(J_{\max})$	Activation energy of $J_{\max}$	43.90	kJ mol^{-1}
$E_a(R_d)$	Activation energy of R_d	46.39	kJ mol^{-1}
$E_a(K_c)$	Activation energy of K_c	79.43	kJ mol^{-1}
$E_a(K_o)$	Activation energy of K_o	36.38	kJ mol^{-1}
$E_a(\Gamma^*)$	Activation energy of Γ^*	37.83	kJ mol^{-1}

Table S3: **Three-grid spatial convergence (GCI) and temporal stationarity verification of the lattice-Boltzmann solver. A**, Celik et al. (2008) Grid Convergence Index across coarse ($\Delta x = 0.288\,\mathrm{mm}$, 128×96), medium ($\Delta x = 0.192\,\mathrm{mm}$, 192×144), and fine ($\Delta x = 0.144\,\mathrm{mm}$, 256×192) grids (`T16_resolution_gci.csv`). **B**, Comparison of legacy 30 k-step (5.2 s) snapshots against 150 k-step (26.0 s) steady-state exports (`T15_stationarity.csv`).

A. Three-grid Richardson extrapolation and Grid Convergence Index (`T16_resolution_gci.csv`)

Gravity	Quantity	Fine (0.144)	Med (0.192)	Coarse (0.288)	Order p	Extrap. ϕ_{ext}	$\mathrm{GCI}_{\mathrm{fine}}$ (%)	$\mathrm{GCI}_{\mathrm{coarse}}$ (%)
$1\,g$	g_{bl} ($\mathrm{mol\,m^{-2}\,s^{-1}}$)	0.9800	0.9860	0.9970	0.75	0.9552	3.16	5.47
$1\,g$	$\Delta C_{\mathrm{CO_2}}$ (mean)	−0.1640	−0.1630	−0.1610	1.00	−0.1670	2.29	4.49
$0\,g$	g_{bl} ($\mathrm{mol\,m^{-2}\,s^{-1}}$)	0.4394	0.4407	0.4433	1.17	0.4364	0.87	1.97
$0\,g$	$\Delta C_{\mathrm{CO_2}}$ (mean)	−0.3665	−0.3654	−0.3635	0.65	−0.3718	1.81	2.79

B. Temporal stationarity at 150 k steps (26.0 s) versus 30 k steps (`T15_stationarity.csv`)

Scenario	Gravity	g_{bl} (30 k)	g_{bl} (150 k)	Snapshot bias (%)	Drift last 5 s (%)	Stationary
Leaf · Earth	$1\,g$	0.997	0.997	0.0	0.00	Yes
Leaf · μg	$0\,g$	0.494	0.443	+11.5	0.00	Yes
Rosette · Earth	$1\,g$	0.540	0.540	0.0	0.00	Yes
Rosette · μg	$0\,g$	0.291	0.272	+7.0	0.00	Yes
Leaf · μg fan lo	$0\,g$	0.885	0.885	0.0	0.00	Yes
Leaf · μg fan mid	$0\,g$	1.554	1.554	0.0	0.00	Yes

Table S4: **Leaf-shape and blade-curl sensitivity (`T14_shape_sweep.csv`) and spatial distribution of leaf-surface $\mathrm{CO_2}$ drawdown (`T18_surface_distribution.csv`).** In **A**, the longitudinal outline group (52-cell chord) and transverse curl group (27–35-cell chord) are independently controlled.

A. Anatomical leaf outline and blade-curl sweep (`T14_shape_sweep.csv`)

Scenario	Group / Cut	Curl	Gravity	g_{bl} ($\mathrm{mol\,m^{-2}\,s^{-1}}$)	δ (mm)	$\Delta C_{\mathrm{CO_2}}$ mean	$\Delta C_{\mathrm{CO_2}}$ peak	Ratio to ref.
`leaf-earth`	Outline (long.)	0.0	$1\,g$	0.997	0.67	−0.161	−0.247	1.000
`shape-flat-earth`	Outline (long.)	0.0	$1\,g$	0.941	0.71	−0.171	−0.276	0.944
`leaf-ug`	Outline (long.)	0.0	$0\,g$	0.443	1.50	−0.364	−0.414	1.000
`shape-flat-ug`	Outline (long.)	0.0	$0\,g$	0.426	1.56	−0.378	−0.441	0.962
`curl-none-earth`	Curl (transv.)	0.0	$1\,g$	1.248	0.53	−0.129	−0.208	1.000
`curl-strong-earth`	Curl (transv.)	0.8	$1\,g$	1.016	0.66	−0.158	−0.326	0.814
`curl-none-ug`	Curl (transv.)	0.0	$0\,g$	0.543	1.23	−0.297	−0.338	1.000
`curl-strong-ug`	Curl (transv.)	0.8	$0\,g$	0.452	1.47	−0.356	−0.540	0.832

B. Spatial distribution of $|\Delta C_{\mathrm{CO_2}}|$ across surface cells (`T18_surface_distribution.csv`)

Scenario	Hardware	Gravity	Mean	Peak	Surface CoV	p_{10} / p_{50} / p_{90}	p_{90}/p_{10}	Frac. > Earth mean
`leaf-earth`	Open box	$1\,g$	0.161	0.247	0.232	0.121 / 0.156 / 0.224	1.84	0.306
`leaf-ug`	Open box	$0\,g$	0.364	0.414	0.125	0.296 / 0.376 / 0.411	1.39	1.000
`rosette-earth`	Open box	$1\,g$	0.298	0.716	0.489	0.149 / 0.258 / 0.522	3.50	0.832
`rosette-ug`	Open box	$0\,g$	0.593	1.064	0.264	0.414 / 0.550 / 0.825	1.99	1.000
`canopy-earth`	Open box	$1\,g$	0.364	0.638	0.459	0.058 / 0.403 / 0.557	9.69	0.847
`canopy-ug`	Open box	$0\,g$	0.829	1.188	0.368	0.189 / 0.893 / 1.130	5.99	0.907
`hw-veggie` (leaf)	VEGGIE	$0\,g$	0.086	0.129	0.364	0.035 / 0.094 / 0.124	3.49	0.000
`hw-rosette-veggie`	VEGGIE	$0\,g$	0.289	0.910	0.657	0.111 / 0.215 / 0.608	5.50	0.680

Table S5: **Chamber domain-size sensitivity (`T17_domain_size.csv`).** Isolated leaf and rosette at $1\,g$ and $0\,g$ in chambers $1\times$ ($36.9 \times 27.6\,\mathrm{mm}$), $1.5\times$ ($55.3 \times 41.5\,\mathrm{mm}$), and $2\times$ ($73.7 \times 55.3\,\mathrm{mm}$) the production domain at fixed physical leaf size and lattice resolution ($\Delta x = 0.288\,\mathrm{mm}$).

Scale	Gravity	Domain factor	Grid ($n_x \times n_y$)	Box (mm)	g_{bl} ($\mathrm{mol\,m^{-2}\,s^{-1}}$)	δ (mm)	Ratio to $1\times$	Drift last 5 s (%)
Leaf	$1\,g$	1.0×	128×96	36.9×27.6	0.997	0.67	1.000	0.00
Leaf	$1\,g$	1.5×	192×144	55.3×41.5	1.015	0.66	1.018	0.00
Leaf	$1\,g$	2.0×	256×192	73.7×55.3	1.037	0.64	1.040	0.00
Leaf	$0\,g$	1.0×	128×96	36.9×27.6	0.443	1.50	1.000	0.00
Leaf	$0\,g$	1.5×	192×144	55.3×41.5	0.344	1.94	0.777	0.00
Leaf	$0\,g$	2.0×	256×192	73.7×55.3	0.297	2.24	0.670	0.00
Rosette	$1\,g$	1.0×	128×96	36.9×27.6	0.540	1.23	1.000	0.00
Rosette	$1\,g$	1.5×	192×144	55.3×41.5	0.501	1.33	0.928	0.20
Rosette	$1\,g$	2.0×	256×192	73.7×55.3	0.505	1.32	0.935	−0.20
Rosette	$0\,g$	1.0×	128×96	36.9×27.6	0.272	2.45	1.000	0.00
Rosette	$0\,g$	1.5×	192×144	55.3×41.5	0.194	3.43	0.713	0.00
Rosette	$0\,g$	2.0×	256×192	73.7×55.3	0.161	4.12	0.592	0.00

Table S6: **Real BRIC and CARA Petri-dish cross-section simulations (`T19_dish_hardware.csv`) and transient CO_2/humidity-feedback dish summaries (`T20_dish_feedback_summary.csv`).** A leaf 1 mm above the agar floor inside actual BRIC (60 mm round, 53.2 mm interior width, 7.38 mm mid-headspace) and CARA (120 mm square, 9.79 mm mid-headspace) cross-sections.

A. Real dish cross-sections without feedback (`T19_dish_hardware.csv`)

Dish geometry	Boundary	Gravity	Headspace (mm)	Grid ($n_x \times n_y$)	g_{bl} vs. cabin	g_{bl} vs. dish air	ΔC_{CO_2} surf.
BRIC-mid (60 mm)	Open to cabin	1 g	7.38	187×48	0.247	0.285	−0.652
BRIC-mid (60 mm)	Open to cabin	0 g	7.38	187×48	0.197	0.237	−0.818
BRIC-mid (60 mm)	Sealed	1 g	7.38	187×48	0.070	0.330	−2.297
BRIC-mid (60 mm)	Sealed	0 g	7.38	187×48	0.060	0.185	−2.679
CARA-mid (120 mm)	Open to cabin	1 g	9.79	419×54	0.251	0.265	−0.642
CARA-mid (120 mm)	Open to cabin	0 g	9.79	419×54	0.176	0.190	−0.916
CARA-mid (120 mm)	Micropore tape	1 g	9.79	419×54	0.247	0.263	−0.653
CARA-mid (120 mm)	Micropore tape	0 g	9.79	419×54	0.170	0.186	−0.945

B. Transient dish simulations with CO_2-limited photosynthesis and humidity-limited transpiration (`T20`)

Run ID	Boundary	Gravity	Duration (s)	End A (%)	$t(A < 10\%)$ (s)	g_{bl} at 5 s	g_{bl} at 26 s
`BRIC_sealed_fb_g1`	Sealed	1 g	415.2	0.3	203.0	0.314	0.287
`BRIC_sealed_fb_g0`	Sealed	0 g	415.2	0.3	207.5	0.224	0.186
`CARA_taped_fb_g1`	Micropore tape	1 g	51.9	95.6	Not reached	0.273	0.266
`CARA_taped_fb_g0`	Micropore tape	0 g	51.9	93.7	Not reached	0.210	0.192

Table S7: **Zero-dimensional enclosure drawdown and 12 h carbon-gain sensitivity across headspace volume, leaf area, tape permeance, irradiance, and cabin CO_2 (`T15_enclosure_sensitivity.tsv`).**

Hardware class	V (mL)	A_{leaf} (cm^2)	Q ($\mu mol\,m^{-2}\,s^{-1}$)	k_{seal} ($mol\,m^{-2}\,s^{-1}$)	Cabin CO_2 (ppm)	t_{comp} (min)	12 h carbon gain (% Earth)
Sealed (BRIC)	15	1.0	60 / 1000	0.0	400	62.9 / 29.2	2.93 / 1.33
Sealed (BRIC)	15	3.0	60 / 1000	0.0	400	20.9 / 9.7	0.98 / 0.44
Sealed (BRIC)	15	10.0	60 / 1000	0.0	400	6.2 / 2.8	0.29 / 0.13
Sealed (BRIC)	30	1.0	60 / 1000	0.0	400	125.9 / 58.5	5.86 / 2.65
Sealed (BRIC)	30	3.0	60 / 1000	0.0	400	41.9 / 19.4	1.95 / 0.88
Sealed (BRIC)	30	6.0	60 / 1000	0.0	400	20.9 / 9.7	0.98 / 0.44
Sealed (BRIC)	30	10.0	60 / 1000	0.0	400	12.5 / 5.8	0.59 / 0.27
Sealed (BRIC)	60	1.0	60 / 1000	0.0	400	252.0 / 117.1	11.72 / 5.30
Sealed (BRIC)	60	3.0	60 / 1000	0.0	400	83.9 / 39.0	3.91 / 1.77
Sealed (BRIC)	60	10.0	60 / 1000	0.0	400	25.1 / 11.6	1.17 / 0.53
Tape (CARA)	30	3.0	60 / 1000	0.001	400	Bounded	75.95 / 54.33
Tape (CARA)	30	3.0	60 / 1000	0.003	400	Bounded	91.36 / 81.07
Tape (CARA)	30	3.0	60 / 1000	0.006	400	Bounded	95.46 / 89.88
Tape (CARA)	30	3.0	60 / 1000	0.012	400	Bounded	97.52 / 94.46
Tape (CARA)	30	3.0	60 / 1000	0.020	400	Bounded	98.35 / 96.30
Sealed (BRIC)	30	3.0	60	0.0	800 / 1200	73.3 / 98.1	4.17 / 6.40
Tape (CARA)	30	3.0	60	0.006	800 / 1200	Bounded	98.67 / 99.37
Ventilated (VEGGIE)	30	3.0	60	∞	400 / 800 / 1200	Bounded	99.58 / 99.88 / 99.94

Table S8: **NASA OSDR metabolomics census: all twelve metabolite-profiling studies (`T03_osdr_metabolomics_census.tsv`).** Queried across all 567 NASA OSDR accessions (`scripts/osdr_census.py`); demonstrates that 0 of 12 metabolomics studies are of a plant.

Accession	Organism	Study title / focus
OSD-108	*Mus musculus*	Muscle atrophy and metabolic alteration in spaceflight
OSD-116	*Mus musculus*	Spaceflight-induced changes in liver metabolism
OSD-138	*Mus musculus*	Multi-omics of rodent eye and retina following flight
OSD-145	*Homo sapiens*	Human astronaut serum metabolomics (NASA Twin Study)
OSD-343	*Mus musculus*	Cecal metabolome and microbiome in microgravity
OSD-360	*Rattus norvegicus*	Rat musculoskeletal metabolomics
OSD-417	*Mus musculus*	Rodent spleen and bone marrow metabolic profiles
OSD-571	Microbial consortium	ISS environmental microbiome and metabolome
OSD-734	*Mus musculus*	Murine cardiovascular and plasma metabolomics
OSD-832	*Homo sapiens*	Commercial astronaut plasma metabolomic profiling
OSD-833	*Homo sapiens*	Spaceflight urine metabolomics time series
OSD-834	*Homo sapiens*	Astronaut peripheral blood mononuclear cell metabolome

Table S9: **Specification and physiological rationale for the 21 predicted metabolite pools (`T06_predicted_compounds.tsv`).** Quasi-steady-state flux-ratio propagation ($P_{\mathrm{flt}}/P_{\mathrm{gc}} = J_{\mathrm{flt}}/J_{\mathrm{gc}}$). Every value is model output, never measured.

Compound	KEGG ID	Driver	Tier	Physiological basis / rationale
2-Phosphoglycolate	C00988	V_o	1	Primary product of Rubisco oxygenase activity
Glycolate	C00160	V_o	1	Dephosphorylated 2-PG exported to peroxisome
Glyoxylate	C00048	V_o	1	Peroxisomal oxidation of glycolate by GOX
Glycine	C00037	V_o	1	Peroxisomal transamination product; mitochondrial GDC substrate
L-Serine	C00065	V_o	1	Mitochondrial condensation product of 2 Gly by GDC/SHMT
Hydroxypyruvate	C00168	V_o	1	Peroxisomal transamination product of L-serine
D-Glycerate	C00258	V_o	1	Reduced hydroxypyruvate; chloroplastic GLYK substrate
Ammonia	C00014	V_o	1	Stoichiometric mitochondrial release from GDC reaction
3-Phosphoglycerate	C00197	A	1	Primary stable product of Rubisco carboxylation
Sucrose	C00089	A	1	Primary transport carbohydrate synthesized from triose-P
D-Glucose	C00031	A	1	Free hexose pool downstream of sucrose cleavage
D-Fructose	C00095	A	1	Free hexose pool downstream of invertase activity
Maltose	C00208	A	1	Transitory starch breakdown and export disaccharide
Starch	C00369	A	1	Chloroplastic insoluble reserve from net assimilation
D-Ribulose 1,5-bisphosphate	C01182	$-C_c$	2	Accumulates upstream of CO_2-limited Rubisco
L-Asparagine	C00152	$-A$	2	Canonical dark/carbon-starvation marker (*ASN1/DIN6*)
L-Valine	C00183	$-A$	2	Branched-chain amino acid mobilized during carbon starvation
L-Leucine	C00123	$-A$	2	Branched-chain amino acid catabolized for alternative respiration
L-Isoleucine	C00407	$-A$	2	Branched-chain amino acid catabolized for alternative respiration
L-Glutamine	C00064	$-A$	2	Nitrogen remobilization hub during carbon deficit
L-Glutamate	C00025	$-A$	2	Primary amino donor for transamination and GS/GOGAT hub

Table S10: **PaintOmics 4 pathway integration control excluding the predicted metabolite layer (`T17_integration_control_nometa.tsv`).** Comparison of combined Fisher p-values using only empirical OSD-522 RNA-seq + Proteomics versus trimodal integration with the 21-compound metabolite scaffold (`job1_oK2h66740Q`). Top 10 pathways plus all evaluated carbon/photosynthesis pathways are shown.

Database	Pathway	p_{RNA}	p_{Prot}	p_{comb} (no metab.)	p_{comb} (with metab.)	FDR (no metab.)
KEGG	Protein processing in ER (UPR)	1.60×10^{-8}	0.0238	8.65×10^{-9}	8.65×10^{-9}	2.01×10^{-6}
MapMan	Cellular response overview	5.75×10^{-5}	0.0024	2.33×10^{-6}	2.33×10^{-6}	2.70×10^{-4}
MapMan	Large enzyme families	1.08×10^{-4}	0.1224	1.61×10^{-4}	1.61×10^{-4}	0.0122
MapMan	Biotic stress	2.57×10^{-5}	0.6841	2.10×10^{-4}	2.10×10^{-4}	0.0122
KEGG	Glucosinolate biosynthesis	0.0962	3.51×10^{-4}	3.82×10^{-4}	6.02×10^{-4}	0.0177
KEGG	Plant hormone signal transduction	1.24×10^{-4}	0.8496	0.0011	0.0011	0.0414
MapMan	Transport overview	0.0191	0.0096	0.0018	0.0018	0.0516
MapMan	Overview	0.5137	4.30×10^{-4}	0.0021	0.0021	0.0516
KEGG	Porphyrin metabolism	0.4048	6.44×10^{-4}	0.0024	0.0101	0.0516
KEGG	Metabolic pathways	3.33×10^{-4}	0.8322	0.0025	0.0119	0.0516
KEGG	Photosynthesis (`ath00195`)	0.1069	0.0094	0.0080	0.0080	0.1423
KEGG	Photosynthesis – antenna (`ath00196`)	0.8153	0.0040	0.0217	0.0217	0.2292
KEGG	Glyoxylate & dicarboxylate (`ath00630`)	0.7194	0.9618	0.9468	0.9937	1.0000

Table S11: **Independent pathway enrichment and Mann–Whitney shift tests outside PaintOmics (`T16_independent_pathways.tsv`).** Direct Fisher exact over-representation tests (FDR $<$ 0.05 DEGs) and two-sided Mann–Whitney U tests on OSD-522 shoot RNA-seq against KEGG REST pathway definitions (`scripts/independent_pathway_test.py`).

KEGG ID	Pathway name	Universe N	DEGs	Fisher OR	Fisher p	Median $\log_2$ FC	Shift	Mann–Whitney p
`ath00195`	Photosynthesis	45	9	2.68	0.0127	−0.245	−0.268	2.17×10^{-10}
`ath00196`	Photosynthesis – antenna proteins	21	1	0.53	0.8472	−0.125	−0.147	1.96×10^{-3}
`ath00710`	Carbon fixation in photosynthetic org.	66	3	0.51	0.9294	−0.082	−0.105	1.32×10^{-4}
`ath00500`	Starch and sucrose metabolism	142	19	1.66	0.0339	−0.142	−0.165	3.51×10^{-6}
`ath00630`	Glyoxylate and dicarboxylate metab.	76	5	0.75	0.7894	−0.102	−0.125	3.15×10^{-4}
`ath04141`	Protein processing in ER (UPR)	203	40	2.66	4.86×10^{-7}	−0.020	−0.043	0.1078

Supplementary Figures

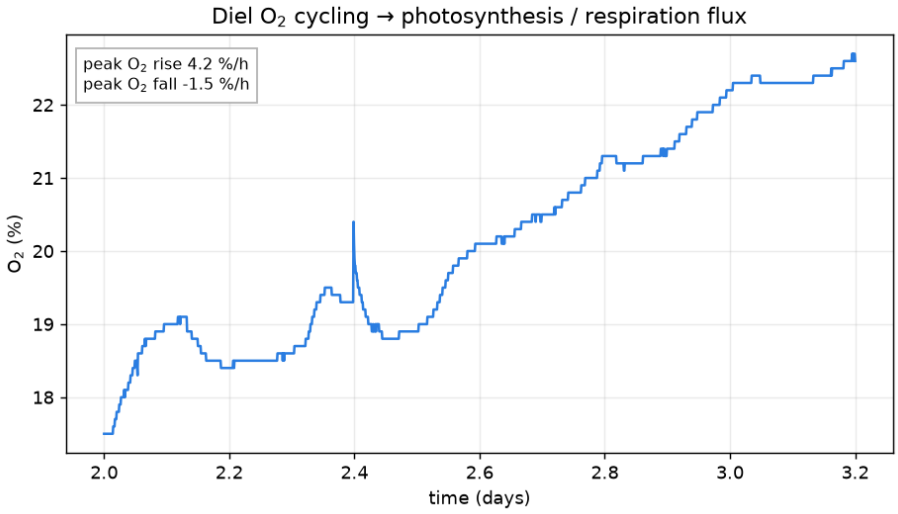


Figure S1: **Single diel O_2 cycle and whole-chamber flux extraction.** Expanded 24 h window from the 4.8 d Vernier closed-chamber trace (`F2_diel_flux.png`) showing linear regression slopes for peak light-phase net O_2 evolution ($+4.2\%\,h^{-1}$) and dark-phase respiratory O_2 uptake ($-1.5\%\,h^{-1}$).

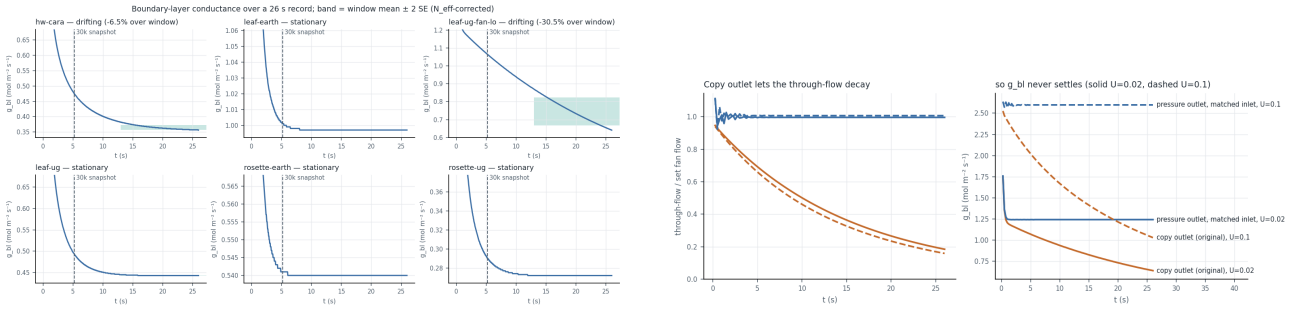


Figure S2: **Temporal stationarity verification and density-matched pressure-outlet validation.** **a**, Time evolution of boundary-layer conductance $g_{bl}(t)$ over 26 s (150 k steps; `F11_stationarity.png`) across six representative scenarios. The vertical dashed line marks the legacy 30 k-step (5.2 s) snapshot, which caught the $0\,g$ pure-diffusion runs prior to full relaxation. **b**, Forced-airflow outlet throughput (Q_{out}/Q_{set}) and $g_{bl}(t)$ comparing the legacy zero-gradient copy outlet against the pressure-pinned outlet ($\rho = 1$) paired with a density-matched velocity inlet (`F12_outlet_bc.png`), which delivers 99.6–100.9% of the set flow.

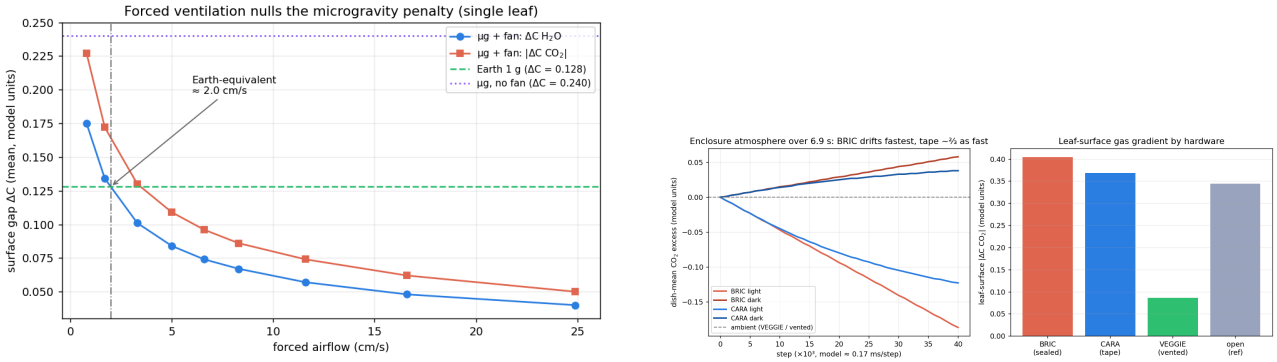


Figure S3: **Single-leaf forced-airflow sweep and spaceflight hardware enclosure comparison. a**, Steady-state surface H_2O and $|CO_2|$ concentration gaps versus fan speed for an isolated microgravity leaf (`F6_forced_airflow.png`), crossing the $1\,g$ free-convection reference at $\approx$ 2.0–2.1 cm s^{-1}. **b**, Short-window (6.9 s) lattice-Boltzmann enclosure-mean CO_2 drift and leaf-surface concentration gradient across BRIC (sealed), CARA (micropore tape), and VEGGIE (ventilated) boundary conditions (`F7_hardware_compare.png`).

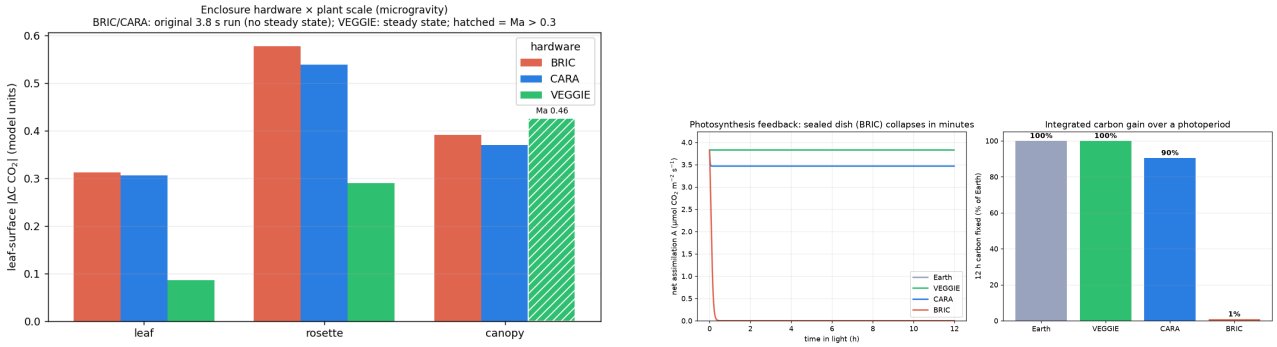


Figure S4: **Hardware boundary conditions across plant scales and closed-loop CO_2-limited photosynthesis. a**, Leaf-surface concentration gaps across BRIC, CARA, and VEGGIE for single leaf, rosette, and microgreen canopy scales in microgravity (`F9_hardware_by_scale.png`). BRIC and CARA have nearly identical surface boundary layers at every scale; one VEGGIE fan speed ($\sim$ 8 cm s^{-1}) restores the leaf and rosette mean but exceeds the low-Mach limit in the dense canopy (hatched bar, Ma = 0.46). **b**, Coupled 0-D enclosure mass balance and CO_2-limited photosynthesis over a 12 h photoperiod (`F10_photosynthesis_feedback.png`), showing collapse of net assimilation within minutes in sealed BRIC (1% of Earth carbon gain) versus 90% in taped CARA and 100% in ventilated VEGGIE.

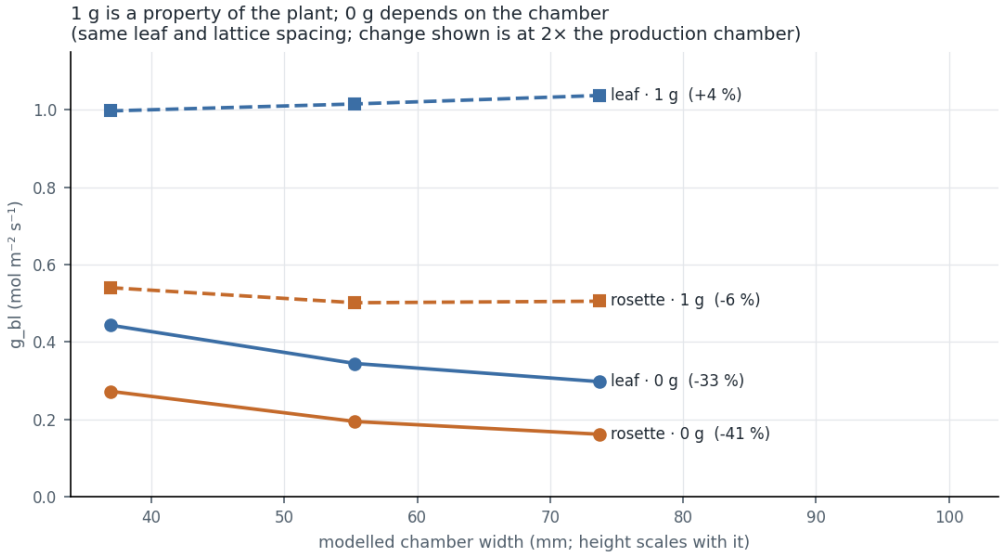


Figure S5: **Chamber domain-size sensitivity at $1\,g$ and $0\,g$.** Boundary-layer conductance g_{bl} across 1×, 1.5×, and 2× domain sizes for an isolated leaf and a rosette (`F13_domain_size.png`, Table S5). At $1\,g$, buoyant convection localizes the boundary layer and g_{bl} changes by $\leq 7\%$; at $0\,g$, 2-D pure diffusion spans the chamber to the ambient walls, so larger domains yield even lower $0\,g$ conductances (-33% leaf, -41% rosette at 2×).

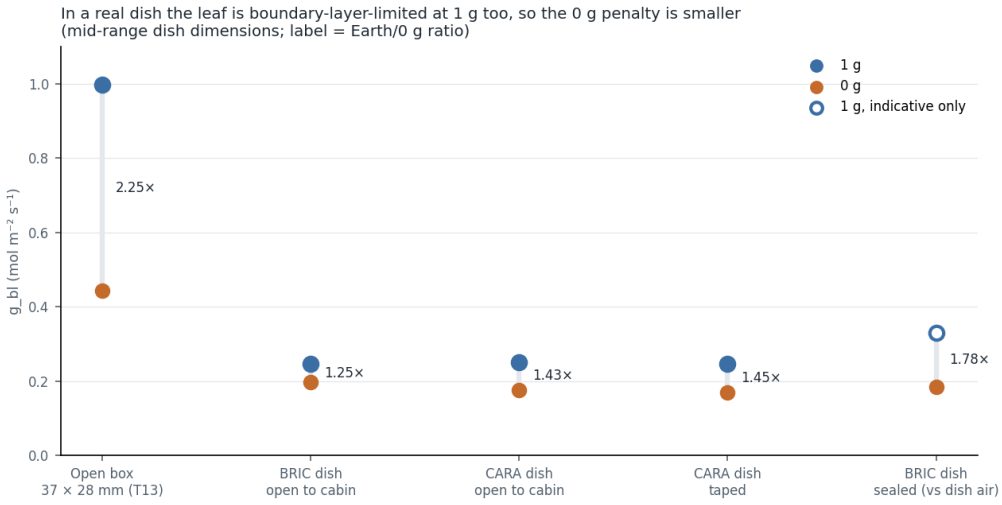


Figure S6: **Leaf boundary-layer conductance inside real BRIC (60 mm round) and CARA (120 mm square) Petri-dish cross-sections.** Simulated g_{bl} at $1\,g$ and $0\,g$ for a leaf positioned 1 mm above the agar floor inside actual dish geometries (`F14_dish_hardware.png`, Table S6A) compared against the generic open chamber. Because the shallow 7.4–9.8 mm headspace restricts vertical plume development even at $1\,g$, dish $g_{bl}(1\,g)$ is $\approx 0.25\,\mathrm{mol\,m^{-2}\,s^{-1}}$ and the $1\,g/0\,g$ ratio inside the dish is 1.25–1.8×.

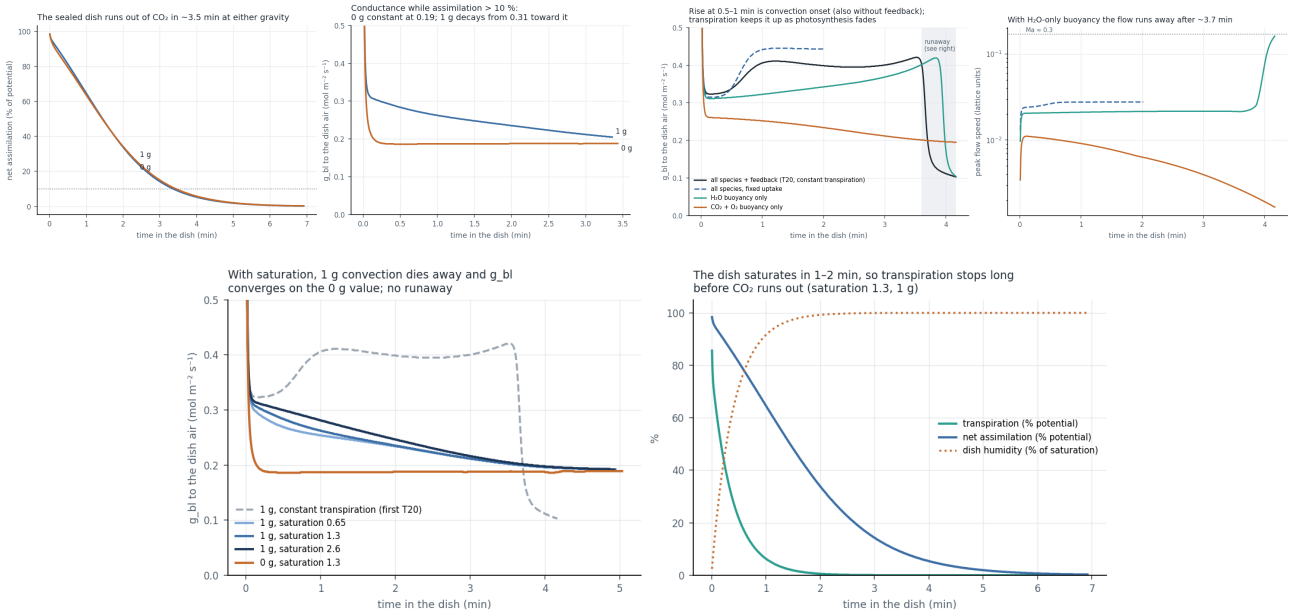
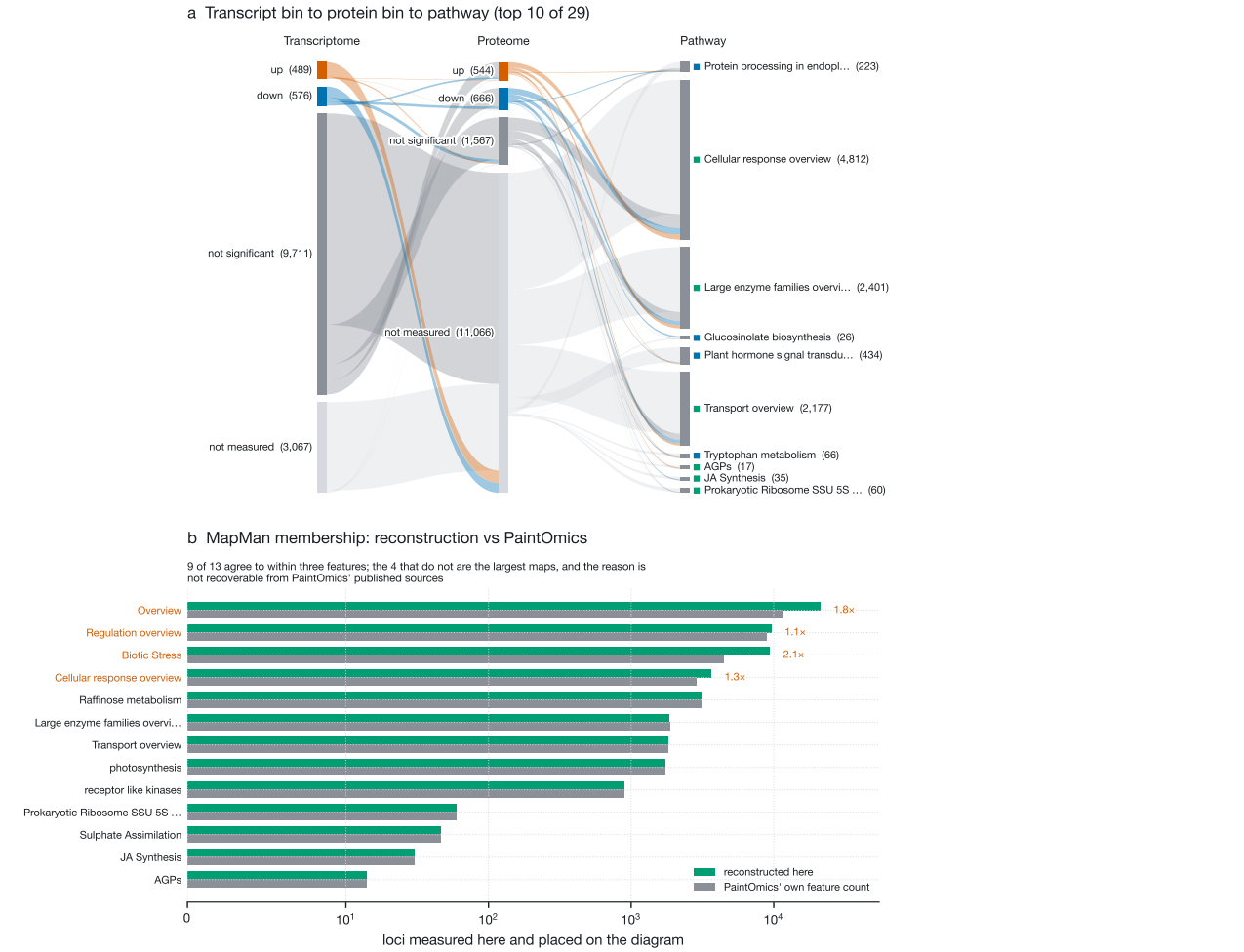


Figure S7: **Transient CO_2 feedback, solutal buoyancy decomposition, and humidity-limited transpiration inside a sealed BRIC Petri dish. Top left** (`F15_dish_feedback.png`), Net assimilation (% of potential) and leaf-to-dish g_{bl} over 415 s in a sealed BRIC dish at $1\,g$ and $0\,g$ (`T20_dish_feedback.csv`): CO_2 is exhausted in ~ 3.5 min ($A < 10\%$ at ≈ 205 s) at both gravities. **Top right** (`F16_dish_buoyancy_diag.png`), Buoyancy species decomposition at $1\,g$ (`T21_dish_buoyancy_diag.csv`), showing that unbounded constant transpiration in a sealed dish causes late-time humidity runaway after ~ 3.7 min. **Bottom** (`F17_dish_humidity.png`), Coupling transpiration to headspace humidity saturation (`T22_dish_humidity.csv`, `h2oSatExcess` $= 0.65, 1.3, 2.6$) eliminates the runaway: as the dish air saturates within 1–2 min, transpiration ceases, $1\,g$ solutal convection dies out, and the $1\,g$ and $0\,g$ boundary-layer conductances converge.

Supplementary Figure S1 — transcript, protein and pathway together



Ribbon width is a locus count. Columns 1 and 2 conserve loci exactly; column 3 counts memberships. Three whole-ontology MapMan maps are excluded (each covers more than 25% of measured loci). Blue square = KEGG, green = MapMan, as in Fig. 6a.

Figure S8: **Multi-omics locus conservation across significant pathways and deterministic MapMan reconstruction verification.** **a**, Sankey diagram (`figS1_sankey.pdf`) tracing every measured *Arabidopsis* locus in the 29 non-broad significant pathways ($p < 0.05$) from its OSD-522 transcript bin (column 1) to its protein bin (column 2) and into the top ten enriched KEGG and MapMan pathways (column 3). Columns 1 and 2 conserve gene counts exactly (13,843 loci); column 3 counts pathway memberships (22,147). **b**, Verification of reconstructed MapMan pathway feature counts (`scripts/mapman.py`, built from GoMapMan Araport11 mappings and MapManStore diagram XMLs) against the feature counts reported by the PaintOmics 4 server: 9 of 13 pathways agree within ≤ 3 features.